

# SSC CGL (COMBINED GRADUATE LEVEL - INSPECTOR, ASO, TAX ASST) — SUBJECT COMPLETE QUESTION BANK

PRACTICE MATERIAL • VERIFIED QUESTION BANK

Total Questions: 243 • Mode: Single Subject Comprehensive Practice

- Q.1.** '©@'%@ 'YG 'éB")M'ò .y 20% 'Y@ "YC'iM'y? "TMK'%G '©0 •ù 'yC"TM#" "2 '©(" '©\$ 'éG•" "ù\$'%G '©M" \$"ù6'B " 'é@ 'Y0"  
A) 16.67% (16 !T%) B) 20%  
C) 25% D) 15%
- Q.2.** •ù 'iA'Y>'%&"é0 'Y?'%@ "Y8"ù\$" \* " # R " R " &"2 "Ù0'é?'R ") (Successive Discounts) 'iG'I> "TMH-B 'Y2 "%0.'IA")M'ò , 'ùM"  
A) 28% B) 30%  
C) 25% D) 27%
- Q.3.** •ù 'ÉG•%0."é( "YM'ù>'©>" @ •Y\*'%@ "Y8"ù\$" • " "2 "Ù0'ò .")2"Ù/ '©0 'ÉG'©("r "â &"é5"â " \$"â 9", \* " 'IA 1 'Y?'yM" > 'YG "%0M"  
A) 11.11% (11 !Q%) B) 10%  
C) 12.5% D) 9%
- Q.4.** 'YK•, "%00"é6"ò 8"é"é0'2 , "Ù/"é 'Y@ 'i0 "%0G 8 "Y0"Ù7"1 'éG•" &"1 " (" 9"2 "é\$" 9"%0d "Y9" "%00"é6"ò "ù\$'%G "Y0"Ù7"1 'éG•"  
A) 24 "Y0"Ù7 / 24 Years B) 16 "Y0"Ù7 / 16 Years  
C) 32 "Y0"Ù7 / 32 Years D) 20 "Y0"Ù7 / 20 Years
- Q.5.** 110,000 'Y@ " >"i? '©0 10% "Y>" M"y?'R & " 8"r " 5" M"r "r 'YM" 5"9&"ù"ò , "Ù/"é 'i0 "%0>'y>" # 'ÉM'ù>'Â "â •)\$" „F-f'erence  
A) 1100 B) 1150  
C) 1200 D) 150
- Q.6.** 'ù&"ò B = 2 : 3 'l%"â " C = 4 : 5 "TMK, 'IK A : B : C 'Y> •Y(" \*é\$ 'YM'ù> "TMK'y>?  
A) 8 : 12 : 15 B) 6 : 9 : 15  
C) 8 : 10 : 15 D) 4 : 6 : 10
- Q.7.** A 'Y?'%@ 'Y>" M'ò "2 &"ù( 'éG•" \$'Y> B •TM8" "é0"Ù/ 'YK 15 'i?', .y '©B" > 'Y0 "%0 'l> "TMH-B &"1("1 'é?) " " , "é0"Ù/ 'YK 'Y"  
A) 6 'i?', ò b F ys B) 8 'i?', ò , F ys  
C) 5 'i?', ò R F ys D) 7 'i?', ò r F ys
- Q.8.** '©>•Y\* A •ù 'ù 'Y@ 'YK 6 '%0 'ùG 'éG•" -" 8'Y\$"â 9", " \*é '¢ " "%0G 8 '%0 'ùG 'éG•" "é2" " 8'Y\$"â 9"%0d 'iK'%0K•" \*é '¢ 'R 8"é"  
A) 24 '%0 'ùG / 24 Hours B) 14 '%0 'ùG / 14 Hours  
C) 18 '%0 'ùG / 18 Hours D) 12 '%0 'ùG / 12 Hours
- Q.9.** 180 'é@'ù0 ") 'É@ •ù " G") "é!"É@ 54 'Y?'é@/'%0 'ù> 'Y@ '©>" " 8"r 'R •)-"r "2 "ù\$'%G "%0G'Y ' .y '©>" " G'y@?  
A) 12 "%0G'Y ' ò " 6V6öæG0 B) 10 "%0G'Y ' ò 6V6öæG0  
C) 15 "%0G'Y ' ò R 6V6öæG0 D) 18 "%0G'Y ' ò , 6V6öæG0
- Q.10.** "i>•)\$ 'É2 'éG•" 'R ("é5 'Y@ '©>" " "ù." ù •) "â 9", \$'Y> 'y>" > 'Y@ '©>" " "ù." ù •) "â 9"%0d 'y>" > 'YG •Y(" ")2 (Downstream)  
A) 3 '%0 'ùG / 3 Hours B) 4 '%0 'ùG / 4 Hours  
C) 3.6 '%0 'ùG / 3.6 Hours D) 4.5 '%0 'ùG / 4.5 Hours
- Q.11.** '©M" %"â S \* "Ù0"é "9\$"ù "%0 'iM'ù>'9 'Y> 'l8'B "Ù/"â 9"1 "ãð  
A) 25.5 B) 25  
C) 26 D) 50
- Q.12.** '©?'l> 'i0 '©A'IM" " 5" M'l."é( •i/" "â '%0A'©>'B R 2 "TMH-B B 5" M"r , "é& •TM('Y@ •i/" "â /"1 43 "Y0"Ù7 "TMK'y>-B \*ù\$"â " "  
A) 25 "Y0"Ù7 / 25 Years B) 30 "Y0"Ù7 / 30 Years  
C) 35 "Y0"Ù7 / 35 Years D) 20 "Y0"Ù7 / 20 Years
- Q.13.** 'ù&"ò 8•) "Ù/"â C f 3cs2 ("Â f" 8"r \*)0"Ù#I "Y?'Ù>'ÉM'ò 9"2Â \$"2 v2r "r 8"Ù%"é( '©0 'YL', 8"â •) •i 'y>?  
A) 7 B) 2  
C) 5 D) 6
- Q.14.** 'yA'9('12 (7 y u - 3 u x) 'éG•" 'Y>•, "â •) (Unit Digit) 'YM'ù> "TMK'y>?  
A) 4 B) 6  
C) 3 D) 7

- Q.15.** 'iK "% 'iM'ù>'9 'Y> •Y(" \*é\$ 3 : 4 "™H 'l%"â '% "â .-i8-b ,,,4b' B 9"%d •™('Y> ")f"%f (LCM) 'YM'ù> "™K'y>?  
A) 48 B) 36  
C) 24 D) 60
- Q.16.** 140 '©M" \$"ò "ù "Ù0"â 5"é2"r "é5"" "2 "c \*™Ù0'I? 'Y?'yM" > "Y>")G '©>"Y2 'YG "%>'R "ù8 •Y(" \*é\$ 'éG•" .ù2"é/"â "é 'Y? 'é  
A) 3 : 1 B) 1 : 3  
C) 2 : 1 D) 3 : 2
- Q.17.** •ù "YC'IM'B " \* " ?'y? 44 "%G'é@ "™H-B "% "â "Ù7"y\$"Ù0'12 (Area) 'YM'ù> "™K'y>?  
A) 154 "%G'é@<sup>2</sup> / 154 cm<sup>2</sup> B) 308 "%G'é@<sup>2</sup> / 308 cm<sup>2</sup>  
C) 77 "%G'é@<sup>2</sup> / 77 cm<sup>2</sup> D) 616 "%G'é@<sup>2</sup> / 616 cm<sup>2</sup>
- Q.18.** •ù " ) 'Â 5"9\$"Ù\$ / "i 'YA 'YG •i"é0 'Y@ 'IM" ?'ÉM'ù> 7 "%G'é@ 'l%"â •) "é 24 "%G'é@ "™H-B 6•) " "â 5'YM" \*97"Ù " / 'YM  
A) 550 "%G'é@<sup>2</sup> / 550 cm<sup>2</sup> B) 500 "%G'é@<sup>2</sup> / 500 cm<sup>2</sup>  
C) 616 "%G'é@<sup>2</sup> / 616 cm<sup>2</sup> D) 450 "%G'é@<sup>2</sup> / 450 cm<sup>2</sup>
- Q.19.** A 'lO B "%G 'YM" .i 120,000 'lO 130,000 ") "é " 'R 5"Ù"é\*é0 "iA" B 'Y?'ù>-B 5" M"r "r •)\$ 'éG•" " 2 115,000 'Y> ")>'Ò 9" -  
A) 19,000 B) 16,000  
C) 17,500 D) 110,000
- Q.20.** 'ù&"ò , 2 ÷, Ò B 9"2Â \$"2 <" 2 ÷<" "â . "é( 'YM'ù> "™K'y>?  
A) 14 B) 16  
C) 18 D) 12
- Q.21.** 'ù&"ò F â , = 4/3 "™K, 'lK sin ,; , "â . "é( 'YM'ù> "™K'y>?  
A) 4/5 B) 3/5  
C) 5/4 D) 3/4
- Q.22.** "YM'ù 'É 24 ÷ 4 × (3 + 1) - 6 + 2 'Y> "%0")\$'â . "é( 'YM'ù> "™K'y>?  
A) 20 B) 18  
C) 24 D) 16
- Q.23.** 'iK "YM'ù "Ù\$ "ò 'R 9" 8"Ù%"é( "%G 'YM" .i 4 'Y?'é@/'% "ù> 'lO 5 'Y?'é@/'% "ù> 'Y@ 'y\$ "ò 8"r 5"ù\* " @'B &"ù6"é •" .y '©2'lO  
A) 27 'Y?'é@ / 27 km B) 15 'Y?'é@ / 15 km  
C) 3 'Y?'é@ / 3 km D) 30 'Y?'é@ / 30 km
- Q.24.** •ù '©A"%>"R .y 'iK •™."Ù." &"Y>" %"yd 'É@'l("r 5"é2"r 'ÉM'é@'i5"é0 'YK 'YA"" 5"% 'é\$"1 'Y> 60% 'é\$ 'é?")> 'lO "Y9 1200 '  
A) 6,000 B) 5,000  
C) 7,200 D) 4,800
- Q.25.** A 'lO B 'Y?'%>@ 'Y>" M'ò "2 "Ù0'é6•2 b &"ù( 'lO 8 'i?', .y 'Y0 "% "lG "™H•)d '©B" G 'Y>" M'ò " . 'É&")0" " C 9"%d A 'Y> "™  
A) 1800 B) 1600  
C) 1700 D) 1750
- Q.26.** 'ù&"ò 'R 5"9\$"Ù\$ 'lO •ù "Y0"Ù 'Y> '©0"ù. "é\* "% . "é( "™K, 'lK •™('YG 'YM"yG'IM" +")K•" "â '%A'©>'B "Ù/"â 9"1 "ãð  
A) 14 : 11 ("YC'IM'B "â , '<"â' ò B 11 B) 11 : 14  
C) 22 : 7 D) 1 : 1
- Q.27.** 'Y?'%>@ "% 'iM'ù> 'YK 56 "%G 'Ù>'r &"y("r \* " 6"y7"12 29 '©M" >'©M'B 9"1\$ "â 9"%d •™8" 8•) "Ù/"â "2 , 8"r -"é 'iG'%G '©0 "iG  
A) 5 B) 3  
C) 7 D) 1
- Q.28.** 1500 'Y@ 'y(" >"i? '©0 2 "Y0"Ù7 'Y> "%>'y>" # 'ÉM'ù>Â "C 9"%d "Y>" M"y?'R , "Ù/"é 'lO 'YM'ù> "™H?  
A) 4% B) 5%  
C) 6% D) 3.5%
- Q.29.** •ù 'É2"Ù2"y, "é '%G •Y\*"%>@ 11"Y@•" \*é0" .y 100 " ( 'É("é 'É?"%8"r "% "â "%\$ 5 " ( 'É""Â 'ù>-B 5" '©>" @ 'YG 'É>'b  
A) 50 " ( / 50 Runs B) 45 " ( / 45 Runs  
C) 55 " ( / 55 Runs D) 60 " ( / 60 Runs
- Q.30.** 9 'lO 25 'Y> 'é"Ù/"é(" \*é\$" „ÖV â &÷ ÷ tional) 'YM'ù> "™K'y>?  
A) 15 B) 17  
C) 12 D) 20

- Q.31.** A 'Y?'%@ 'Y>' M'd '2 ' &"ù('1 'éG•" " " %@@ 'Y>' M'd '2 , &"ù('1 'éG•" \*)0"â " 8'Y\$"â 9"%d "YG 4 'i?'%K•" \$'R 'R 8"é%  
A) 6 'i?'€ B) 8 'i?', f, F ys)  
C) 10 'i?'€ D) 7 'i?'€
- Q.32.** 180 'é@'ù0 ") 'É@ •ù " G") "é!"É@ 54 'Y?'é@/'% 'ù> 'Y@ 'y\$"ò 8'r "" 0"TM@ "TMH-B 'R •)-"r '2 \*"é0 'Y0"%G 'éG•" /"" "ù\$'%  
A) 10 "%G'Y ' B) 12 "%G'Y ' f " 6V6öæG2•  
C) 15 "%G'Y ' D) 18 "%G'Y '
- Q.33.** '15,000 'Y@ 'y(" >"i? '©0 2 "Y0"Û7 'YG ")?•ò R 5"é0"Û7"ù 'i0 "%G '© "Û0"YC'iM'y? 'ÉM'ù>'Â " 8"é"é0'2 , "Û/"é 'Y> •Y 'I0 'Y  
A) '120 B) '150 ( '150)  
C) '180 D) '200
- Q.34.** •ù 'iA'Y>'%&"é0 'Y?'%@@ "Y8"Û\$" "r •) "ù\$ 'éB")M'd \*" # R " ") 'iG'%G 'YG 'É>'b -" # R "â 2"é- •Y0"Û "ù\$ 'Y0'I> "TMH-B /  
A) '650 B) '700  
C) '750 ( '750) D) '800
- Q.35.** •ù 'YH")G 'éG•" " Â S \*"%8"r " #R \*"%8"r "r 8"ù "Û"r 2 4 : 5 'YG •Y(" \*"é\$ 'éG•" 9"% -B 'i? 'YH")G 'éG•" " 2 'y(" >"i? '170  
A) 80 B) 100  
C) 120 (120 coins) D) 150
- Q.36.** •ù " 'Â 5"9\$"Û\$" / 'ÉG") ( 'YG •i"é0 'Y@ 'iM" ?'ÉM'ù> 7 "%G'é@ 'I0 •© '©>•, 8"y." 9"%d •y8'Y> 'YA"" \*"97"Û " / 'YM"yG'iM" •  
A) 648 "%G'é@<sup>2</sup> B) 748 "%G'é@<sup>2</sup> (748 cm<sup>2</sup>)  
C) 848 "%G'é@<sup>2</sup> D) 924 "%G'é@<sup>2</sup>
- Q.37.** 'ù&"ò 6-â , + cos ;, Ò !£" 6÷2 , 'IK cos ;, Ò 6-â , 'Y> 'é>', "Û/"â 9"1 "ãð  
A) " 2 sin ;€ B) " 2 tan ;€  
C) sin ;€ D) 1
- Q.38.** 'ù&"ò 8'éA'©M'©/ A '©0 "% 'É 'r " 8"Û5'IA")M'd ...&Vf/ÆPxive), "%."é?'B ...7-ÖÖWG ic) 'I%"â 8•) "Û0"é.'R ...@ransitive) 'I@'%  
A) " ?'YM'B 8•,•) / Empty Relation B) 'IA")M'ù\$"â 8•,•) / Equivalence Relation  
C) '©M" \$"ù8'é."ù\$ "% 'É 'r ò çF"×7-ÖÖWG ic Relation D) "%>" M"Y\$"Û0"ù "% 'É 'r ò Væ—`ersal Relation
- Q.39.** 'ù&"ò "+"( f: R !' R, f(x) = 2x 'iM"Y>" > '©0"ù-"é7"ù\$ "TMH, 'IK f 'Y?" , \*"Û0'Y>" "â +")( "TMH?  
A) •ù "% " \$'Y> •i "Û "é&'R ò öæRÖöæR æB öçFò „&-|V7F—'e)B) 'É9" Û 'R \$'Y> •i "Û "é&'R ò Ö ày-one and Onto  
C) •ù "% " \*" 'IA •Y("é "Û "é&'R ò öæRÖöæR ut not Onto D) ', \$"2 'YH'Y@ ', 9" '©M'1>'i / Neither One-one nor Onto
- Q.40.** 'éA'iM'ò ."é( "i>'i> (Principal Value Branch) 'éG•" 6-â »'f ó" "â ."é( 'YM'ù> "TMK'y>?  
A) < ób f3 • B) < ó2 fç •  
C) < ób fC[ • D) < ó' f" •
- Q.41.** tan {1(1) + cos {1(-1/2) + sin {1(-1/2) 'Y> 'é>', "Û/"â 9"1 "ãð  
A) < ó@ B) < ó  
C) 3< ó@ D) <
- Q.42.** sin {1(sin(2< ó2" "â 8"TM@ 'é>', "Û/"â 9"1 "ãð  
A) 2< ó0 B) < ó0  
C) -< ó0 D) 4< ó0
- Q.43.** 'ù&"ò "YM'ùB"" Ò ¶ ö-¥Ò 'R 8'é."ù\$ •i5"Û"/)9 (Symmetric Matrix) "TMH, 'IK 'YL', 8" 6" M'B 8'iM'ò 9"1 " ð  
A) a\_ij = -a\_ji B) a\_ij = a\_ji  
C) a\_ij = 0 D) a\_ij = 1
- Q.44.** 'ù&"ò " " 8'é>', "1 "ò "r 5" M'r "YM'ùB"" 9"% , 'IK (AB)' 'Y> 'é>', "ù8'YG 'É0"é," 9"1\$"â 9" fð  
A) A'B' B) B'A'  
C) AB D) BA
- Q.45.** 'ù&"ò 'R 2 r 2 "1 "ò "â 5"Û/" \$"Û "Û0'é#" / (Invertible) •i5"Û"/)9 "TMH, 'IK |adj(A)| 'Y> 'é>', "ù8'YG 'É0"é," 9"1 "ãð  
A) |A| B) |A|<sup>2</sup>  
C) |A|<sup>3</sup> D) 3|A|
- Q.46.** 'ù&"ò "ù8" 8"é0'9?'R " "1 'iK '© 'YM'I?'ù>• ...&ðws) •Y%"Y> 'iK "%M'I 'Ò „6öÇVÖç2' 8" M"Y8'â „-FVçF-6 Â' 9"1 , 'IK "%>" #  
A) 1 B) 0 / Zero  
C) -1 D) •Y\*“ ?'é?'B ò -æf-æ—FP

- Q.47.** •ù “Y0”Ù •i5”Ù/)9 A “YM’ùA’IM’YM” .’9@’ò „-àvertible) ‘Y9’>’I> “™H ’ù&“ò “ ”y5”” /i?:  
 A)  $|A| = 0$  B)  $|A| \neq 0$  (•Y5”Ù/” \$”Ù ”Ù0’é#” / ’%9” , Non-singular)  
 C)  $A = A'$  D)  $A = -A'$
- Q.48.** ’12, ’ Ò ÆÖr†6-â , ’ “â , ”r 8“é\*y ”Ù7 •Y5’Y2’Â †G’öG, ’ ”Ù/“â 9”1 “ãð  
 A)  $\tan x$  B)  $\cot x$   
 C)  $-\cot x$  D)  $\sec x$
- Q.49.** ’ù&“ò ’ Ò F â »’,f7, Ò <2’ ò f Ò 7<”” 9”2Â \$”2 G’öG, “â .“é( ‘YM’ù> “™K’y>?  
 A)  $3 / (1 + x^2)$  B)  $1 / (1 + x^2)$   
 C)  $3 / (1 - x^2)$  D)  $2 / (1 + x^2)$
- Q.50.** ’ù&“ò , Ò 6÷2 , ’l%“â ’ Ò 6-â , “™K, ’IK dy/dx ‘Y> ’é>, ”Ù/“â 9”1 “ãð  
 A)  $\tan ;\text{€}$  B)  $-\cot ;\text{€}$   
 C)  $\cot ;\text{€}$  D)  $-\tan ;\text{€}$
- Q.51.** “YC’IM’B ”r ”Ù7”y\$”Ù0’12 ’éG•” \$”Ù0”ù ”Ù/“â ” ”r 8“é\*y ”Ù7 ’©0”ù5“ M’l( ‘Y@ ’i0 ‘YM’ù> “™K’y@ ’É, r = 5 cm “™K?  
 A)  $10 < 6\text{Ü}''\text{ö6Đ}$  B)  $5 < 6\text{Ü}''\text{ö6Đ}$   
 C)  $25 < 6\text{Ü}''\text{ö6Đ}$  D)  $20 < 6\text{Ü}''\text{ö6Đ}$
- Q.52.** “Y ”Ù0  $y = x^3 - x$  ‘YG ’É?•)&” , Ò ” \*” 8”Ù\*” M“b 0”y “â ” \*”Ù0“Y#’I> (Slope of Tangent) ‘YM’ù> “™K’y@?  
 A) 11 B) 12  
 C) 10 D) 6
- Q.53.** ”+  $(1 / (1 + x^2)) dx$  ‘Y> ’é>, ”Ù/“â 9”1\$“â 9”fð  
 A)  $\sin \{^1 x + C$  B)  $\tan \{^1 x + C$   
 C)  $\log(1 + x^2) + C$  D)  $\cos \{^1 x + C$
- Q.54.** ”+  $\sec x dx$  ‘Y> “%.”é “)( ’é>, ”Ù/“â 9”1 “ãð  
 A)  $\log |\sec x + \tan x| + C$  B)  $\log |\sec x - \tan x| + C$   
 C)  $\sec x \tan x + C$  D)  $\tan^2 x + C$
- Q.55.** ”+ e.2 †6-â , ^2 6÷2 , ’ G, “â .“é( ‘YM’ù> “™K’y>?  
 A) e.2 6÷2 , ^2 0 B) e.2 6-â , ^2 0  
 C) -e.2 6-â , ^2 0 D) e.2 †6-â , Ò 6÷2 , ’ ^2 0
- Q.56.** ’%?”iM’©?’B 8’é>’Y2’, ”2 âfÀ/2)  $(\sin x / (\sin x + \cos x)) dx$  ‘Y> ’é>, ”Ù/“â 9”1 “ãð  
 A) < ó B) < ó@  
 C) 1 D) 0
- Q.57.** ’ù&“ò b†,’ ’R 5“ù7’â +“( (Odd Function,  $f(-x) = -f(x)$ ) “™H, ’IK ”+ < • C  $f(x) dx$  ‘Y> ’é>, ”Ù/“â 9”1 “ãð  
 A)  $2 \text{ ”+ € C f(x) dx}$  B) 0 / Zero  
 C) a D) 1
- Q.58.** •Y5’Y2 “%.” “ #  $(d^2y/dx^2)^3 + (dy/dx)^2 + \sin(dy/dx) + 1 = 0$  ‘Y@ ’%>’B „FVpree) ‘YM’ù> “™H?  
 A) 3 B) 2  
 C) 1 D) ’©0“ù-“é7“ù\$ ’%9” / Not Defined
- Q.59.** “ H’i?’R “Y ”” 8’é@‘Y0’2 G’öG, ^2 ’ Ò “â 8’é>’Y2’, ” #’R „-çFVprating Factor - I.F.) ‘YM’ù> “™K’I> “™H?  
 A)  $e^{(+ P dx)}$  B)  $e^{(+ Q dx)}$   
 C)  $+ P dx$  D)  $e^{(- + P dx)}$
- Q.60.** •Y5’Y2 “%.” “ #  $dy/dx = y/x$  ‘Y> “YM’ù>’© “™2 (General Solution) ‘YM’ù> “™K’y>?  
 A)  $y = Cx$  B)  $y = C / x$   
 C)  $y = x^2 + C$  D)  $xy = C$
- Q.61.** ’ù&“ò &”2 6”)(”Ù/”y\$”Ù\$” 8’i?’iK•” r “ ”r ”r 2”ù a × . b x = 0 “™K, ’IK •™(‘YG ’É@’¢ “â ”1# , , ”Ù/“â 9”1 “ãð  
 A) 0° B) 45°  
 C) 90° (< ó”Â \*” 8”Ù\*” 2•), “Y\$ / Perpendicular) D) 180°
- Q.62.** i0” r †E × k0”” “â .“é( ‘YM’ù> “™K’y>?  
 A) 0 B) 1  
 C) -1 D) k0

**Q.63.** “%&“ù6 a x = 2i0" ² 6£ + 6k0" “â \* “ ?’é>’2 „Ö væ—GVFR Æ wÂ’ “ù\$’%> “™K’y>?

- A) 7  
C) 49
- B) 11  
D) " 41

**Q.64.** 'ù&"ò rr"rÒ r9"2Â\$"\$8'i?"b r " " r"r8),.)' 'éG•" "Ù/"â 8'IM"ò 9"fđ

- A)  $\vec{a} \times \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \times \vec{b}$  / Perpendicular  
 B)  $\vec{a} \times \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \times \vec{b}$  (Parallel / Collinear)  
 C)  $|\vec{a} \times \vec{b}| = |\vec{b} \times \vec{a}|$   
 D)  $\vec{a} \times \vec{b} = \vec{b} \times \vec{a}$

**Q.65.** 'ù&"ò 'R 0"y "â " &"ù "ÒÙ "¹ "Ù"/é • „F—&V7F-öâ 6÷6-æW2' ÂÂ ÒÂ â 9"¹ , 'IK 'YL' , 8"â 8•),•)' "%&"%5 "%\$ "Ù/ "™H?

- A)  $l + m + n = 1$   
B)  $l^2 + m^2 + n^2 = 1$   
C)  $l^2 + m^2 + n^2 = 0$   
D)  $l^2 - m^2 + n^2 = 1$

**Q.66.** 'É?•)&" ‡,Â •, z) 'Y@ 'éB"" , "ù 'iA (Origin - 0, 0, 0) "%oG 'iB" @ 'YM'ù> "™K'y@?

- A)  $x + y + z$  B) " $(x^2 + y^2 + z^2)$ "  
C)  $x^2 + y^2 + z^2$  D) " $(x + y + z)$ "

**Q.67.** 'iK " G'i>•ù 'É?'%o "r &"ù "ÒÙ '%oA'©>'B "Ù0'é6•2 Â " Â 2 \$'Y> a ,, b ,, c , "™H•"Â \*" 8"Ù\*" 2), "Y\$ (Perpendicular) "™K•) "

- A)  $a \bullet a = b \bullet b = c \bullet c$ ,  
 B)  $a \bullet a + b \bullet b + c \bullet c = 0$   
 C)  $a \bullet a + b \bullet b + c \bullet c = 1$   
 D)  $a \bullet b - a \bullet b = 0$

**Q.68.** “ H‘i?‘R \*”Ù0”1 ”Ù0“é.’, 8‘é8”Ù‘“â „Å ’.”y •™&”Ù&”y6”Ù/ ’12’, „ö&|V7F—`e Function) Z = ax + by ‘Y> •y7”Ù ‘l. ’é>’, „÷ F–Ö Â Val

- A) "%A"% "y\$ "YM"yG"IM" "r 6" 0"U7 ('YK'9@'ò' ,"ù 'iA'9 "©0 / Center point of feasible region only at origin  
 B) "%A"% "y\$ "YM"yG"IM" "r 6" 0"U7 ('YK'9@'ò' ,"ù 'iA'9 "©0 / Center point of feasible region  
 C) "%A"% "y\$ "YM"yG"IM" "r " "YG.&"U0 "©0 / Exact center  
 D) 'YM"yG"IM" "r "u8" -" /é&"9"U "ù 'É?&)" \*" ò ày random point

**Q.69.** 'ù&"ò " " &"2 8"Ù5'I 'IM" 'ù("é • „-æFW VæFVçB W`ents) "™K•"Â \$"2 „ " " " "â .é( 'YM'ù> "™K'y>?

- A)  $P(A) + P(B)$       B)  $P(A) \times P(B)$   
C)  $P(A) / P(B)$       D)  $P(A) - P(B)$

**Q.70.** 'ù&"ò „' Ò 2ó,Â „'" Ò ó" \$'Y> P(A ") B) = 1/4 "™K, 'IK P(A | B) 'Y> 'é>', "Ù/"â 9"¹ "ãð

- A)  $\frac{1}{2}$  B)  $\frac{2}{3}$   
C)  $\frac{3}{4}$  D)  $\frac{1}{8}$

**Q.71.** 'Y?'"%@ '% '">%> E 'Y@ '©M" >'ù?'Y\$"â „R' \$'Y> •™8'Y@ '©B" '% '">%> P(E) 'Y> 'ùK'r "Ù/"â 9"1\$"â 9" fđ

- A) 0  
B) 1  
C) 0.5  
D) Indeterminate

**Q.72.** 'ù&"ò 'R 5" M'r "YM'ùB"" 9",Â \$"2 2 r 8'iH"R "ù8 '©M" "é0 'Y> •i5'Ù"/)9 "TMK'I> "TMH?

- A) "Y?"y.-"%o.é?B "YM'ùB"" ò 6°ew-symmetric B) "%o.é?B "YM'ùB"" ò 7-ÖÖWG ic Matrix  
C) •y "é •i5"Ù"/)9 / Identity Matrix D) "iB'%"M"ò "YM'ùB"" ò cVÆÄ Ö G ix

**Q.73.** '12', b±, ' Ò <2 Ò 7, "â 8"Ù%"é(" / •™ "Ù "ù7"Ù (Local Maxima) 'Y?', , "ù 'iA '©0 "™K'y>?

- A)  $x = 1$  B)  $x = -1$   
C)  $x = 0$  D)  $x = 3$

**Q.74.**  $\int \frac{1}{a^2 - x^2} dx$  'Y> 'é>', "Ù/"â 9"1\$ "â 9" f ð

- A)  $\sin \{^1(x/a) + C$  B)  $(1/a) \sin \{^1(x/a) + C$   
C)  $\cos \{^1(x/a) + C$  D)  $\log|x + \sqrt{a^2 - x^2}| + C$

**Q.75.** 'i' 6•2 8'é>'Y2', „çFVpration by Parts) “%B'IM“ "2 R b G, Ò R "2 b G, Ò "2 .Rr, "2 b G,•Ò G, .y '©M“ %â +“( '©A'‰(“r “â 8“™

- A) BODMAS %?%?ù. B) ILATE %?%?ù. (Inverse, Logarithmic, Algebraic, Trigonometric, Exponential)

**Q.76.** 2 tan {<sup>1</sup> x 'Y> 'é>', 6-â »' "r \*'iK•". }y 'YM'ù> "TM'K'l> "TMH?

- A)  $\sin \{1(2x / (1 + x^2))\}$  B)  $\sin \{1(2x / (1 - x^2))\}$   
C)  $\sin \{1((1 - x^2) / (1 + x^2))\}$  D)  $\sin \{1(x / (1 + x^2))\}$

**Q.77.** 'ù&"ò "YM'ùB"" " "1 "ò " r 2 \$'Y> B 'Y@ 'YK'ù? 3 × 4 "™K, 'IK 'yA'9('12 •i5"ù"/)9 AB 'Y@ 'YK'ù? (Order) 'YM'ù> "™K'y@?

- A)  $3 \times 3$  B)  $2 \times 4$   
C)  $4 \times 2$  D) 'yA'9('‰ 'Û5 '‰9" "™H / Multiplication not possible

**Q.78.**  $x \cdot Y = |Y| \cos \theta$  (Direction Cosines of x-axis)

- A) (1, 0, 0) B) (0, 1, 0)  
C) (0, 0, 1) D) (1, 1, 1)

- Q.79.**  $f(x) = \begin{cases} x^2 \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$  is continuous at  $x = 0$  but not differentiable at  $x = 0$ .  
 A) Continuous and Differentiable at  $x = 0$   
 B) Continuous but not Differentiable at  $x = 0$   
 C) Neither Continuous nor Differentiable at  $x = 0$   
 D) Discontinuous at  $x = 0$
- Q.80.**  $\int_0^1 \frac{1}{1+x^2} dx$  is equal to  
 A)  $\frac{\pi}{4}$   
 B)  $\frac{\pi}{2}$   
 C)  $\frac{\pi}{6}$   
 D)  $\frac{\pi}{3}$
- Q.81.** The area of the region bounded by the lines  $x = 0$ ,  $y = 0$ ,  $x + y = 1$  is  
 A)  $\frac{1}{2}$   
 B)  $\frac{1}{4}$   
 C)  $\frac{1}{6}$   
 D)  $\frac{1}{3}$
- Q.82.** The value of  $\int_0^1 x^2 dx$  is  
 A)  $\frac{1}{3}$   
 B)  $\frac{1}{2}$   
 C)  $\frac{1}{4}$   
 D)  $\frac{1}{6}$
- Q.83.** The value of  $\int_0^1 x^3 dx$  is  
 A)  $\frac{1}{4}$   
 B)  $\frac{1}{5}$   
 C)  $\frac{1}{6}$   
 D)  $\frac{1}{7}$
- Q.84.** The value of  $\int_0^1 x^4 dx$  is  
 A)  $\frac{1}{5}$   
 B)  $\frac{1}{6}$   
 C)  $\frac{1}{7}$   
 D)  $\frac{1}{8}$
- Q.85.** The value of  $\int_0^1 x^5 dx$  is  
 A)  $\frac{1}{6}$   
 B)  $\frac{1}{7}$   
 C)  $\frac{1}{8}$   
 D)  $\frac{1}{9}$
- Q.86.** The value of  $\int_0^1 x^6 dx$  is  
 A)  $\frac{1}{7}$   
 B)  $\frac{1}{8}$   
 C)  $\frac{1}{9}$   
 D)  $\frac{1}{10}$
- Q.87.** The value of  $\int_0^1 x^7 dx$  is  
 A)  $\frac{1}{8}$   
 B)  $\frac{1}{9}$   
 C)  $\frac{1}{10}$   
 D)  $\frac{1}{11}$
- Q.88.** The value of  $\int_0^1 x^8 dx$  is  
 A)  $\frac{1}{9}$   
 B)  $\frac{1}{10}$   
 C)  $\frac{1}{11}$   
 D)  $\frac{1}{12}$
- Q.89.** The value of  $\int_0^1 x^9 dx$  is  
 A)  $\frac{1}{10}$   
 B)  $\frac{1}{11}$   
 C)  $\frac{1}{12}$   
 D)  $\frac{1}{13}$
- Q.90.** The value of  $\int_0^1 x^{10} dx$  is  
 A)  $\frac{1}{11}$   
 B)  $\frac{1}{12}$   
 C)  $\frac{1}{13}$   
 D)  $\frac{1}{14}$
- Q.91.** The value of  $\int_0^1 x^{11} dx$  is  
 A)  $\frac{1}{12}$   
 B)  $\frac{1}{13}$   
 C)  $\frac{1}{14}$   
 D)  $\frac{1}{15}$
- Q.92.** The value of  $\int_0^1 x^{12} dx$  is  
 A)  $\frac{1}{13}$   
 B)  $\frac{1}{14}$   
 C)  $\frac{1}{15}$   
 D)  $\frac{1}{16}$
- Q.93.** The value of  $\int_0^1 x^{13} dx$  is  
 A)  $\frac{1}{14}$   
 B)  $\frac{1}{15}$   
 C)  $\frac{1}{16}$   
 D)  $\frac{1}{17}$
- Q.94.** The value of  $\int_0^1 x^{14} dx$  is  
 A)  $\frac{1}{15}$   
 B)  $\frac{1}{16}$   
 C)  $\frac{1}{17}$   
 D)  $\frac{1}{18}$

Q.95. 'ù&"ò 8'é@'Y0'2 'k' Ò ., 2 " Ò "r.)2 "%."é( (Equal) "TMK•"Â \$"2 2 "â ."é( 'YM'ù> "TMK'y>?

- A)  $\pm 4$  B)  $\pm 2$   
C)  $\pm 8$  D) 0

Q.96. A.P. 3, 8, 13, 18, ... 'Y> 10"Y>• \*'b "Ù/"â 9"1 "ãð

- A) 48 B) 50  
C) 45 D) 53

Q.97. '©M" %'â ", \*)0"Ù#"é 'YK•" f 2 " 2 2 2 âââ + 100) 'Y> 'ùK'y+"" "Ù/"â 9"1 "ãð

- A) 5050 B) 5000  
C) 5150 D) 5500

Q.98. A.P. 7, 13, 19, ..., 205 'éG•" " 2 'Y?'I("r \*'b 9"%o ?

- A) 34 B) 33  
C) 35 D) 32

Q.99. %<sup>3</sup>ABC 'éG•" DR "R \$2 9"%od 'ù&"ò B Ò ãR 6ÒÂ D" Ò 2 6ÒÂ R Ò 6Ò 9"2Â \$"2 T2 "â ."é( 'YM'ù> "TMK'y>?

- A) 2 cm B) 1.5 cm  
C) 2.5 cm D) 3 cm

Q.100. 'iK "%." B'¢ \$"Ù0"ù-" "1 'Y@ 'ÙA'É>'9 'Y> •Y(" \*é\$ 4 : 9 "TMH-B '%o "r "Ù7"y\$"Ù0'12"1 'Y> •Y(" \*é\$ 'YM'ù> "TMK'y>?

- A) 16 : 81 B) 2 : 3  
C) 4 : 9 D) 81 : 16

Q.101. •ù "%."YK'2 \$"Ù0"ù-" 'Y> 'Y0"Ù# 13 "%G'é@ 'I0 •ù 'ÙA'É> 5 "%G'é@ "TMH-B \$" 8" @ 'ÙA'É> 'Y@ ")."Ù,"é 'YM'ù> "TMK'y>?

- A) 12 cm B) 10 cm  
C) 11 cm D) 8 cm

Q.102. 'É?'%M'iA'9 (2, 3) 'I0 (4, 1) 'YG 'É@'¢ " &")0" "Ù/"â 9"1 " ð

- A) 2" 2 'é>'IM" / 2" 2 units B) " 8 'é>'IM" / " 8 units  
C) 4 'é>'IM" / 4 units D) 2 'é>'IM" / 2 units

Q.103. 'É?'%M'iA (-6, 8) 'Y@ 'éB"), "ù("Ù&" „÷ igin) "%G 'iB" @ 'YM'ù> "TMK'y@?

- A) 10 'é>'IM" / 10 units B) 14 'é>'IM" / 14 units  
C) 8 'é>'IM" / 8 units D) 6 'é>'IM" / 6 units

Q.104. 'É?'%M'iA'9 (-5, 7) 'I0 (-1, 3) 'YK 'é?"> "%G "Y>")G " G'i>'i ' "r .:yM'òÙ,"ù("Ù&" "r ("ù0"Ù&"y6'é 'R "Ù/"â 9"1 'yG?

- A) (-3, 5) B) (-2, 5)  
C) (3, -5) D) (-3, -5)

Q.105.  $\sin 60^\circ \times \cos 30^\circ + \cos 60^\circ \times \sin 30^\circ$  'Y> 'é>', "Ù/"â 9"1 "ãð

- A) 1 B) 1/2  
C) " 3/2 D) 0

Q.106.  $9 \sec^2 A - 9 \tan^2 A$  'Y> 'é>', "ù8'YG 'É0"é," 9" fð

- A) 9 B) 1  
C) 8 D) 0

Q.107.  $\tan 65^\circ / \cot 25^\circ$  'Y> 'é>', "Ù/"â 9"1 "ãð

- A) 1 B) 0  
C) " 3 D) 1/" 3

Q.108. •ù •©0"Ù""Ù5"é"" •)-"r " \* "é •TM8'Y@ •© '©>•, "r , ">É0 "TMH-B 8")0"Ù/ 'Y> •TM("Ù("ù( 'YK'2 „ ævÆR öb VÆPvation) 'YM'

- A) 45° B) 30°  
C) 60° D) 90°

Q.109. 'ÙB'é? 'YG •ù 'É?•)&" 8"r ." ("é0 'YG '©>'b 8"r 3 ." " &")0" \* " ("é0 'YG "i?'i0 'Y> •TM("Ù("ù( 'YK'2 3 9"%od 'é@'%o>" " •)

- A) 10" 3 m B) 30" 3 m  
C) 15 m D) 20 m

Q.110. "YC'IM'B "r "ù8" ,"é9"Ù/ 'É?•)&" 8"r " '©@ 'y 'iK "%M'©0"Ù6 " G'i>'9 PA 'I0 PB 'Y@ ")."Ù,"é 'ùK•" .:y 'YM'ù> "% 'É 'r 9"1

- A) PA = PB ("%. "é( "TMK'I@ "TMH•"• B) PA > PB  
C) PA < PB D) PA = 2PB

- Q.111.** •Y0"Ù"YC'IM'B .y 'É("â "1# (Angle in a semi-circle) 'Y?'I("â 9"1\$"â 9" fđ  
 A) "‰. 'YK'2 f" ' ò & -v±B ævÆR f" • B) 60°  
 C) 180° D) 45°
- Q.112.** 'ù&"ò 8"Ù" M"b 0"y "â A 'IO PB •ù 'iB"%0"r 8"r f "r "1# '©0 'ÙA'Y@ "TMK"Â \$"2 " ðA 'Y> 'é>', "Ù/"â 9"1 "â „ò "y 'iM" 9", "ð  
 A) 50° B) 60°  
 C) 70° D) 80°
- Q.113.** 'IM" ?'ÉM'ù> r "Y>")G "YC'IM'B "r " , \$"Ù0"ù "Ù/'i ' "â "Ù7"y\$"Ù0'12 'YM'ù> "TMK'y> 'É?"‰ "â "1# „, 9" fđ  
 A) (;, ò 3c ' r År² B) (;, ò f ' r År²  
 C) (;, ò 3c ' r #År D) 2< +
- Q.114.** •ù "YC'IM'B " " " ?'y? 'IO 'YM"yG'IM" +"" 8•) "Ù/"é\$"Ù. 'R 0")\* "‰G 'É0"é, " 9"‰ -B " , 5"9\$"Ù\$ 'Y@ 'IM" ?'ÉM'ù> 'YM'ù> "TMK  
 A) 2 'é>'IM" / 2 units B) < . "é\$"Ù0'R ò Å units  
 C) 4 'é>'IM" / 4 units D) 7 'é>'IM" / 7 units
- Q.115.** 'IM" ?'ÉM'ù> r "Y>")G •ù ' K", " M'y "12"r "â 8'éM'©B" M'2 \*97"Ù " / 'YM"yG'IM" +"" "Ù/"â 9"1\$"â 9" fđ  
 A) 3< + B) 2< +  
 C) 4< + D) 2/3 < +0
- Q.116.** 'ù&"ò 'R "12"r " \$"Ù0"ù "Ù/"â &"1 " (" " &" "é , 'IK •TM8'Y> •i/'I( 'Y?'I("r " ("â 9"2 "é 'y>?  
 A) 8 'yA'‰> / 8 times B) 4 'yA'‰> / 4 times  
 C) 2 'yA'‰> / 2 times D) 16 'yA'‰> / 16 times
- Q.117.** "‰. "é( 'IM" ?'ÉM'ù> 'IO "‰. "é( •© '©>•, 5"é2"r 'R , "y2', " 'R 6•) " "r 'ù\$‰K•" "â "‰A'©>'B "Ù/"â 9"1 "ãð  
 A) 3 : 1 B) 1 : 3  
 C) 1 : 1 D) 2 : 1
- Q.118.** 'é>yM'òÂ . "é"Ù/'R " , "TMA") 'YG 'É@'¢ 8"TM@ •i(" -"Y?'R 8•), •) (Empirical Relationship) 'YM'ù> "TMH?  
 A) 'É9" 2'R Ò 2 . "é"Ù/'R Ò " . "é"Ù/ / Mode = 3 Median - 2 Mean B) 'É9" 2'R Ò " . "é"Ù/'R Ò 2 . "é"Ù/ / Mode = 2 Median - 3 Mean  
 C) 3 'É9" 2'R Ò . "é"Ù/'R Ò " . "é"Ù/ D) 'é>yM'ù = 3 'É9" 2'R Ò " . "é"Ù/
- Q.119.** "‰ 'iM'ù>9 2, 4, 6, 8, 10 'Y> "‰. "é 'IO 'é>yM'ò "Ù/"â 9"1 "ãð  
 A) 6 B) 5  
 C) 7 D) 8
- Q.120.** •i 'Y!'ÉK•" rÂ "Â Â BÂ 'Â RÂ , " . "é"Ù/"ù "â „ÖVF- â' "Ù/"â 9"1 "ð  
 A) 7 B) 8  
 C) 6 D) 5
- Q.121.** 'É 'ù( 2, 3, 5, 3, 7, 3, 8, 3, 9 'Y> 'É9" 2'R "Ù/"â 9"1 "ãð  
 A) 3 B) 5  
 C) 7 D) 9
- Q.122.** 'Y?"‰@ •Y8•)-"R 'ù("â „-x ÷ 76- le Event) 'Y@ '©M" >ù?'Y\$"â "Ù/"â 9"1\$" 9" fđ  
 A) 0 B) 1  
 C) 0.5 D) -1
- Q.123.** •ù '©>"‰G 'YK •ù 'É> " +y 'Y("r " " 'Ù>'ÉM'ò 8•) "Ù/"â ... ime Number) •i("r " " "Ù0"é/"ù 'I> 'YM'ù> "TMK'y@?  
 A) 1 / 2 B) 1 / 3  
 C) 1 / 6 D) 2 / 3
- Q.124.** •Y "Ù " \$" 9 '1G•) " •, S" \*'IM'IK•" " \$"é6 'Y@ 'y!"Ù!" .y "‰G •ù '©\$"Ù\$â ("ù "é2"â "é\$"â 9"‰d •ù ")>"" 0•) 'Y> 'É>i6"é9  
 A) 1 / 26 B) 1 / 52  
 C) 1 / 13 D) 2 / 13
- Q.125.** •ù 'Y>" 8"r " '\$R c "ù. " ù •) "â 8"r "é\$" 9", '\$Y> 40 'Y?'é@/'‰ 'ù> "‰G ")L'ù\$" 9"‰d '©B" @ 'ù>'IM" > 'Y@ 'I8'B "é2 'YM'  
 A) 48 'Y?'é@/'‰ 'ù> B) 50 'Y?'é@/'‰ 'ù>  
 C) 45 'Y?'é@/'‰ 'ù> D) 52 'Y?'é@/'‰ 'ù>
- Q.126.** 15,000 'Y@ " >"i? '©0 10% "Y>" M"y?'R &" 8"r " 5" M"r "r 4' " 4' "â •) \$" "Ù/"â 9"1 "ãð  
 A) 150 B) 125  
 C) 1100 D) 175

- Q.127.** A 'Y?'%@ 'Y>' M'ò "² " &"ù( 'éG•" '\$'Y> B 18 'i?', .y '©B" > 'Y0'I> "™H-B &"¹("¹ 'é?") " "ù\$'%G 'i?', .y '©B" > 'Y0'y 'yG?  
 A) 7.2 'i?', ò rã" F ys B) 8 'i?', ò , F ys  
 C) 6 'i?', ò b F ys D) 7.5 'i?', ò rãR F ys
- Q.128.** 'ù&"ò " (" "r.)²"Ù/ 'éG•" #RR " 5"9&"Ù""ò 9"² "é , 'IK 'yC"™?'9@ 'YK 'i\*B .y 'Y?'I("r \*"Ù0'I?"i\$ 'Y@ 'Y." " (" 9"¹ " \$"é "ò  
 A) 20% B) 25%  
 C) 16.66% D) 15%
- Q.129.** 'ùB'YM")?' 5"ù-"é ' , \*"Ù0'ég'ù?'Y> a = bq + r 'éG•" 6"y7"¹2 r 'YG ")?•ò "É( "%@ "i0"Ù\$ "%\$ "Ù/ "™H?  
 A) 0 "d r < b B) 0 < r "d b  
 C) 0 "d r "d b D) r > b
- Q.130.** "%,%G '1K'ù@ •Y-"é "Ù/ "% 'iM'ù> (Smallest Prime Number) 'YL', 8" 9" f ð  
 A) 2 (ù 'é>'IM" 8'â 'Ù>'ÉM'ò' ò " „ðæÇ' W'en Prime) B) 1  
 C) 3 D) 0
- Q.131.** " H'i?'R , "™A'©& (Linear Polynomial) 'Y@ •Y""ù 'I. '%>'B "ù\$'%@ "™K'I@ "™H?  
 A) 1 B) 2  
 C) 0 D) 3
- Q.132.** 'ù&"ò &"Ù5"ù "é\$ 'É9" \*b <" Ò 7, ² " "r 6")("Ù/"é 'R ± 'I0 , " 9"¹ , 'IK 1/; ² ó² 'Y> 'é>', "Ù/"â 9"¹ "ãð  
 A) 3 / 2 B) 2 / 3  
 C) -3 / 2 D) 1
- Q.133.** 'ù&"ò &"² 0"y "é •" 'R , "ù 'iA '©0 '©M" \$"ù "Ù "y& 'Y0'I@ "™H•"Â \$"² 5"" , "ù 'iA 'iK'%K•" 8'é@'Y0'9K•" "â "%8"â 9"" 9"¹ "ãð  
 A) •Y&"Ù5"ù\$ / "™² (Unique Solution) B) •Y(•)\$ •Y("y "™² / Infinitely Many Solutions  
 C) 'YK•, 9"" ("™@•" ò æð 6ðÇWF-ðâ ... arallel Lines) D) 'iK "™² / Two Distinct Solutions
- Q.134.** 'ù&"ò , ² ' Ò B '\$'Y> x - y = 4 "™K, 'IK x 'I0 y 'YG 'é>', "Ù/"â 9"¹ 'yG?  
 A) x = 9, y = 5 B) x = 8, y = 6  
 C) x = 10, y = 4 D) x = 7, y = 7
- Q.135.** 'iM"Y?'%>'B 8'é@'Y0'2 "â ."é('R 0")\* (Standard Form) 'YM'ù> "™K'I> "™H?  
 A) ax² + bx + c = 0 ('É9"é a "" 0) B) ax + b = 0  
 C) ax³ + bx² + cx + d = 0 D) ax² + c = 0
- Q.136.** "%." " # x² - 9 = 0 'YG 'éB"" "Ù/"â 9"¹ 'yG?  
 A) ±3 B) 9  
 C) 3 D) ±9
- Q.137.** A.P. -10, -6, -2, 2, ... 'Y> "%>" M'R •)\$" †B' "Ù/"â 9"¹ f ð  
 A) 4 B) -4  
 C) 2 D) -2
- Q.138.** '©M" %'â â 5"ù7'â \*"Ù0'é "9\$ "% 'iM'ù>'9 'Y> 'ùK'y+"" ... 7VÒ öb f—'7B â öFB æ GW al numbers) 'YM'ù> "™K'I> "™H?  
 A) n² B) n(n+1)  
 C) 2n D) n(n+1)/2
- Q.139.** 'É?'%M'iA (-4, -5) 'Y?' , 'IA" M'Y>•)6 'éG•" 8"Ù%"ù\$ "™H?  
 A) 'IC'I@'ò 'IA" M'Y>•)6 (Third Quadrant) B) '©M" %'â 'IA" M'Y>•)6 / First Quadrant (+, +)  
 C) 'iM"Y?'I@'ò 'IA" M'Y>•)6 / Second Quadrant (-, +) D) '©\$" 0"Ù% '©\$" 0"Ù%"é "b ò `ourth Quadrant (+, -)
- Q.140.** 'ù&"ò \$" ( 'É?'%M'iA "% " G'i@ (Collinear) "™K•"Â \$"² '%8"r , '%("r 5"é2"r \$"Ù0"ù-" 'Y> 'YM'yG'IM" +"" "Ù/"â 9"¹ "ãð  
 A) 0 B) 1  
 C) -1 D) •Y\* " ?'é?'B 0")\* "%G •Y("y / Infinitely Many
- Q.141.** 'IM" ?'ÙA'Â "r 6" 0"Ù7 (x •, y •), (x ,, y ,), (x f, y f) "™K•"Â \$"² "y("Ù&"Ù0'R "r ("ù0"Ù&"y6'é 'R "Ù/"â 9"¹ 'yG?  
 A) ((x •+x ,+x f)/3, (y •+y ,+y f)/3) B) ((x •+x ,)/2, (y •+y ,)/2)  
 C) (x •+x ,+x f, y •+y ,+y f) D) (x •x ,x f, y •y ,y f)
- Q.142.** 'ù&"ò 6-â , = 3/5 "™K, 'IK cos ; , "â ."é( 'YM'ù> "™K'y>?  
 A) 4 / 5 B) 5 / 4  
 C) 3 / 4 D) 4 / 3

Q.143.  $\tan 45^\circ$  'Y> 'é>', "Ù/"â 9"1\$ "â 9" f ð

- A) 1 B) 0  
C) 1/" 3 D) " 3

Q.144.  $1 + \cot^2$  ;, "ù8'YG 'É0"é, " 9"1\$ "â 9" f ð

- A)  $\operatorname{cosec}^2$  ;€ B)  $\sec^2$  ;€  
C)  $\tan^2$  ;€ D)  $\sin^2$  ;€

Q.145. "YC'IM'B "r 'R , "ù 'iA '©0 'Y?'I(" 8"Û\* M"b 0"y "é •" " '©@ 'É> "%o 'l@ "TMH•#ð

- A) 'YG"Y2 1 (Only one) B) 2  
C) •Y(•)\$ / Infinity (" ) D) 0

Q.146. •ù "YC'IM'B " "ù\$"%@ "%o."é 'l0 "%oM'©0"Û6 " G'ï>•ù (Parallel Tangents) "TMK "%o 'l@ "TMH•#ð

- A) 2 B) 4  
C) 1 D) •Y(•)\$ / Infinity (" )

Q.147. 'IM" '?ÉM'ù> r "Y>")G "YC'IM'B "r 'R 'IA" M'Y>)6 (Quadrant) 'Y> 'YM"yG'IM" +"" "Ù/"â 9"1\$ "â 9" f ð

- A)  $1/4 < +$  B)  $1/2 < +$   
C)  $< +$  D)  $2 <$

Q.148. 'IM" '?ÉM'ù>9 R 'l0 r (R > r) "Y>")G 'iK "%o 'YG•)&"Ù0" / "YC'IM'IK•" "r , " 'É("r 5")/ (Ring) 'Y> 'YM"yG'IM" +"" "Ù/"â 9"1\$ "â 9" f ð

- A)  $< \dots + " \ddot{O} + " \bullet$  B)  $< \dots " \ddot{O} ">$   
C)  $2 < \dots " \ddot{O} " \bullet$  D)  $< + " \ddot{O} +$

Q.149. 'IM" '?ÉM'ù> 7 "%oG'é@ 'l% "â •) "é 10 "%oG'é@ "Y>")G ")."Û,"YC'IM'l@'ò , "y2', "â 'ù\$, "Ù/"â 9"1\$ "â ð

- A)  $1540 \text{ "%oG'é@}^3 / 1540 \text{ cm}^3$  B)  $1450 \text{ "%oG'é@}^3 / 1450 \text{ cm}^3$   
C)  $1500 \text{ "%oG'é@}^3 / 1500 \text{ cm}^3$  D)  $770 \text{ "%oG'é@}^3 / 770 \text{ cm}^3$

Q.150. "i 'YA 'Y@ 'l?' M'ù •© '©>•, ...6Æ çB †V–v‡B vÂr' "â 8")\$ "Ù0 'YM'ù> "TMH?

- A) " ( $r^2 + h^2$ ) B)  $r^2 + h^2$   
C) " ( $r^2 - h^2$ ) D)  $r + h$

Q.151. 'ÙA'É> a "Y>")G •ù '%o( (Cube) 'YG 'éA'iM'ò 5"ù " M'2 „F– vöæ Â' " 2'ém'É>•, "Ù/"â 9"1\$ " ð

- A) a" 3 B) a" 2  
C) 3a D) a / " 3

Q.152. "Y0"Û •Y 'l0"é2 10 - 25 'Y> "Y0"Û '©?"TM M', „6Æ 72 Ö k) 'YM'ù> "TMK'y>?

- A) 17.5 B) 15  
C) 20 D) 35

Q.153. •i 'Yl'ÉK•" "Â #RÂ RÂ ,Â 3RÂ # "â \* " ? "%o0 (Range) 'YM'ù> "TMK'y>?

- A) 23 B) 35  
C) 12 D) 20

Q.154. 'Y?'%o@ '%o?'iM'©?'B 'ù("â ...7W&R W`ent) 'Y@ '©M" >'ù?'Y\$ "â "ù\$"%@ "TMK'l@ "TMH?

- A) 1 B) 0  
C) 0.5 D) •Y("ù6"Û "ù\$ / Indeterminate

Q.155. 'ù&"ò „R' Ò ã R 9"2Â \$"2 tR ("TM@•"r ... †æ÷B R" " \*")Ù0"é/"ù 'l> 'YM'ù> "TMK'y@?

- A) 0.95 B) 0.05  
C) 0.50 D) 1.05

Q.156. 'iK "%o?'YM'YK•" "2 'R 8"é% •TM "é2'%oG '©0 'Y. "%oG 'Y. •ù '©?'IM'B „ B ÆV 7B öæR †V B' '%oG 'Y@ '©M" >'ù?'Y\$ "â "Ù/"â 9"1\$ " ð

- A) 3 / 4 B) 1 / 4  
C) 1 / 2 D) 1

Q.157. •ù "Y8"Û\$ " "2 "s# . "y 'ÉG'©("r \* " # R "â 2"é- "TMK'l> "TMH–B 5"%oM'IA 'Y> 'YM" / 'éB")M'ò „6÷7B ice) 'YM'ù> "TMK'y>?

- A) 1600 B) 1580  
C) 1640 D) 1560

Q.158. 120 'é@ "ù0 ") 'É@ "ùM" G', SB "ù." ù •) "â " "é2 "%oG •ù 'i 'ÛG 'YK 'Y?'I("r 8'é/ 'éG•" \* "é0 'Y0"y " ð

- A) 8 "%oG'Y' ò , 6V6öæG0 B) 6 "%oG'Y' ò b 6V6öæG0  
C) 10 "%oG'Y' ò 6V6öæG0 D) 12 "%oG'Y' ò " 6V6öæG0

Q.159. (7 y u - 3 u x) 'Y> •y "é 'Y> •Y 'R "Ù/"â 9"1 "ãð

- A) 4 B) 6  
C) 0 D) 7

Q.160. 'Y?'l("r 8'é/ 'éG•" "1 'y( 5% "Y>" M"y?'R 8"é""é0'2 ,"Ù/"é 'Y@ 'i0 "%G 'iK'yA'%> "TMK 'É>•ù "ãð

- A) 20 "Y0"Ù7 / 20 Years B) 10 "Y0"Ù7 / 10 Years  
C) 15 "Y0"Ù7 / 15 Years D) 25 "Y0"Ù7 / 25 Years

Q.161. •ù "%G.'É>"TMA 'IM" '?ÙA'Â " \*"Ù0'IM'ùG'R -" "â b 8"y." 9"%d •y8'Y> 'YM"yG'IM" +"" "Ù/"â 9"1 "ãð

- A) 9" 3 "%G'é@<sup>2</sup> / 9" 3 cm<sup>2</sup> B) 18" 3 "%G'é@<sup>2</sup> / 18" 3 cm<sup>2</sup>  
C) 36 "%G'é@<sup>2</sup> / 36 cm<sup>2</sup> D) 6" 3 "%G'é@<sup>2</sup> / 6" 3 cm<sup>2</sup>

Q.162. 3/4, 5/6, 7/8, 9/10 'éG•" 8'É8"r , ' <" -"ù("Ù( 'YL', 8" 9" fð

- A) 9 / 10 B) 7 / 8  
C) 5 / 6 D) 3 / 4

Q.163. '©M" %'â S 8'â \*"Ù0"é "9\$ "% 'iM'ù>'9 'Y> 'l8'B "Ù/"â 9"1 "ãð

- A) 51 B) 50  
C) 50.5 D) 100

Q.164. 'ù&"ð %â‡, Ò ' ² %â‡, ² ' Ò 3# 9"²Â \$"² , "â ."é( 'YM'ù> "TMK'y>?

- A) 7 B) 6  
C) 8 D) 5

Q.165. 20% 'l0 10% 'Y@ 'iK 'YM" ."ù '1B'ð ...7V66W76—`e Discounts) 'YG "%G.'lA")M'ð 'Y2 '1B'ð "Ù/"â 9"1 "ð

- A) 28% B) 30%  
C) 25% D) 22%

Q.166. 'i>•)\$ 'É2 'éG•" ("é5 'Y@ '©>"" "ù." ù •) "â \$'Y> 'y>" > 'Y@ '©>"" "ù." ù •) "â 9"%d 'y>" > 'Y@ 'i?'i> 'éG•" "é2 'YM'ù> "TMK

- A) 12 'Y?'é@/'% 'ù> B) 8 'Y?'é@/'% 'ù>  
C) 10 'Y?'é@/'% 'ù> D) 14 'Y?'é@/'% 'ù>

Q.167. '©>•y\* A 'ù 'Y@ 'YK 10 '% 'ùG 'l%"â \*"é '¢ " R •) "r."y 'Ù0'l> "TMH-B &"1("1 'é?") " "ù\$'%G "%G.'ð ."y 'Ù0'y 'yG?

- A) 6 '% 'ùG / 6 Hours B) 8 '% 'ùG / 8 Hours  
C) 5 '% 'ùG / 5 Hours D) 7.5 '% 'ùG / 7.5 Hours

Q.168. [•i 'É@'©@•ù8 '©@'2 \*"Ù0"é0•)-"ù '©0" "Ù7"â \*"Ù0'iM", Ò ,"%G 'Y?•) '©0" "Ù7"â "r 8•)&" M'Ò ."y 'G'ù> •y 'ù0'©M" '?ùG"i( 'i

- A) "%GM'Y?'l? Alpha 'é>'%M'ð 9"€ B) "%GM'Y?'l? Beta "% " 'l0 '©M" >'é>'9?'R 9"" 9"€  
C) 'iK'%K•" 8"Ù%"ù\$"ù/"é •Y\*" M'ù>'©M'B 9"%G D) •i 'Y!'ÉG •Y\*" M'ù>'©M'B 9"%G

Q.169. BODMAS '%>'ù. "%G "TM2 'Y@'É?•ó 24 / 4 + 3 \* (8 - 5)

- A) 15 B) 27  
C) 12 D) 18

Q.170. 'iM"Y?'%>'B 8'é@'Y0'2 ...ã" Ò w, ² " Ò "r.")2 'ÉM'é>'B " "ù -@

- A) 3, 4 B) -3, -4  
C) 2, 6 D) 1, 12

Q.171. A 'Y?'%G@ 'Y>" M'ð "² &"ù( 'éG•" \$'Y> B •TM8"r R &"ù( 'éG•" \*)0"â " \$"â 9"%d 'iK'%K•" ."ù2'Y0 •TM8 'Y>" M'ð "² "ù\$'%G

- A) 5 'i?'€ B) 6 'i?'€  
C) 7.5 'i?'€ D) 8 'i?'€

Q.172. sin 30° + cos 60° 'Y> 'é>', "Ù/"â 9"1 "ãð

- A) 1 B) 1/2  
C) " 3 D) 0

Q.173. '©M" %'â \*"é '¢ 'Ù>'ÉM'ð 8•) "Ù/"é •" "â ."é""Ù/ 'ÉM'é>'B " "ù -@

- A) 5.6 B) 5.2  
C) 4.8 D) 6.0

Q.174. •ù "Y0"Ù 'Y> '©0"ù."é\* 48 "%G'é@ "TMH-B "%G "â "Ù7"y\$"Ù0'12 'YM'ù> "TMK'y>?

- A) 144 "%G'é@<sup>2</sup> B) 196 "%G'é@<sup>2</sup>  
C) 120 "%G'é@<sup>2</sup> D) 169 "%G'é@<sup>2</sup>

Q.175. “%0” “ : (0.3 \* 0.3 + 0.01) / 0.1

- A) 1.0 B) 0.1  
C) 0.9 D) 10.0

Q.176. [•i0•i0'É@ •ù('ù@'©@“%@ “%@'É@'ù@-1 •Y-”Ù/“é8 '©M“ 6”Ù( 10] •Y 'Y '9?'B “ 8•) ”Ù/“é\$”Ù.'R 'Ù?'YM“y.'I> '©0 •i“é0“ù\$  
A) 'é>% “Y?'Y2”Û\* 1 B) '©M“ .“é#“ù\$ “TM2 “Y?'Y2”Û\* 2

- C) '©M“ >'Y.“ù “%?“iM'y>•)\$ 3 D) •Y(”Ù/ 'Y>“ 4

Q.177. 'YK•, “%0“é6“ò 8“é“é0'2 ,”Ù/“é '©0 6 “Y0”Ù7”1 'éG•" 8”Ù5'ù 'Y@ 3 'yA'%@ “TMK 'É>'I@ “TMH–B 5“é0”Ù7“ù 'ÉM'ù>'Â &“ ”  
33.3% 15.0%

- 12.5% 18.0%

Q.178. 'YK•, “%0“é6“ò 8“é“é0'2 ,”Ù/“é '©0 7 “Y0”Ù7”1 'éG•" 8”Ù5'ù 'Y@ 3 'yA'%@ “TMK 'É>'I@ “TMH–B 5“é0”Ù7“ù 'ÉM'ù>'Â &“ ”  
28.6% 15.0%

- 12.5% 18.0%

Q.179. 'YK•, “%0“é6“ò 8“é“é0'2 ,”Ù/“é '©0 8 “Y0”Ù7”1 'éG•" 8”Ù5'ù 'Y@ 3 'yA'%@ “TMK 'É>'I@ “TMH–B 5“é0”Ù7“ù 'ÉM'ù>'Â &“ ”  
25.0% 15.0%

- 12.5% 18.0%

Q.180. 'YK•, “%0“é6“ò 8“é“é0'2 ,”Ù/“é '©0 9 “Y0”Ù7”1 'éG•" 8”Ù5'ù 'Y@ 3 'yA'%@ “TMK 'É>'I@ “TMH–B 5“é0”Ù7“ù 'ÉM'ù>'Â &“ ”  
22.2% 15.0%

- 12.5% 18.0%

Q.181. 'YK•, “%0“é6“ò 8“é“é0'2 ,”Ù/“é '©0 10 “Y0”Ù7”1 'éG•" 8”Ù5'ù 'Y@ 3 'yA'%@ “TMK 'É>'I@ “TMH–B 5“é0”Ù7“ù 'ÉM'ù>'Â &“ ”  
20.0% 15.0%

- 12.5% 18.0%

Q.182. 'YK•, “%0“é6“ò 8“é“é0'2 ,”Ù/“é '©0 11 “Y0”Ù7”1 'éG•" 8”Ù5'ù 'Y@ 3 'yA'%@ “TMK 'É>'I@ “TMH–B 5“é0”Ù7“ù 'ÉM'ù>'Â &“ ”  
18.2% 15.0%

- 12.5% 18.0%

Q.183. 'YK•, “%0“é6“ò 8“é“é0'2 ,”Ù/“é '©0 12 “Y0”Ù7”1 'éG•" 8”Ù5'ù 'Y@ 3 'yA'%@ “TMK 'É>'I@ “TMH–B 5“é0”Ù7“ù 'ÉM'ù>'Â &“ ”  
16.7% 15.0%

- 12.5% 18.0%

Q.184. 'YK•, “%0“é6“ò 8“é“é0'2 ,”Ù/“é '©0 13 “Y0”Ù7”1 'éG•" 8”Ù5'ù 'Y@ 3 'yA'%@ “TMK 'É>'I@ “TMH–B 5“é0”Ù7“ù 'ÉM'ù>'Â &“ ”  
15.4% 15.0%

- 12.5% 18.0%

Q.185. 'YK•, “%0“é6“ò 8“é“é0'2 ,”Ù/“é '©0 14 “Y0”Ù7”1 'éG•" 8”Ù5'ù 'Y@ 3 'yA'%@ “TMK 'É>'I@ “TMH–B 5“é0”Ù7“ù 'ÉM'ù>'Â &“ ”  
14.3% 15.0%

- 12.5% 18.0%

Q.186. 'YK•, “%0“é6“ò 8“é“é0'2 ,”Ù/“é '©0 5 “Y0”Ù7”1 'éG•" 8”Ù5'ù 'Y@ 3 'yA'%@ “TMK 'É>'I@ “TMH–B 5“é0”Ù7“ù 'ÉM'ù>'Â &“ ”  
40.0% 15.0%

- 12.5% 18.0%

Q.187. 24, 36 'I0 60 'Y> 'é9'IM'I. “%.“é\*“Y0”Ù\$‘R ,,,4b’ ”Ù/“â 9”fð

- A) 6 B) 12  
C) 18 D) 24

Q.188. 'ù&“ò “ù8” 8•) ”Ù/“â “â # R -“é 80 “TMH, 'IK •TM8 “% ‘iM'ù> ‘Y> 35% ‘YM'ù> “TMK'y>?

- A) 120 B) 140  
C) 160 D) 180

Q.189. •ù “Y8”Ù\$” 0” S .”y ‘i0” &‘Y0 “ A 600 ‘éG•” ,”y ” “é\$” 9”%d “>’Ò \*”Ù0'I?“i\$ ‘YM'ù> “TMH?

- A) 15% B) 20%  
C) 25% D) 10%

Q.190. 'ù&“ò B = 2 : 3 'I%“â ” C = 4 : 5 “TMK, 'IK A : B : C ‘ÉM'é>'B ” “ù –@

- A) 8 : 12 : 15 B) 6 : 9 : 10  
C) 2 : 7 : 5 D) 8 : 10 : 15

**Q.191.** “ A 2,000 ‘©0 5% “Y>“ M“y?’R &“ 8”r 2 5“ M“r “â 8“é““é0’2 ,”Ù/“é ‘YM’ù> “TMK’y>?

- A) “ A 250 B) “ A 300  
C) “ A 350 D) “ A 400

**Q.192.** •ù ‘ùM“ G’, B •) ”r .”y 360 ‘Y?’é@ ‘Y@ ‘iB“ @ ‘I/ ‘Y0’I@ “TMH–B “% ” “é2 ‘é@/“%G ‘éG•" ”Ù/“â 9”f ð

- A) 20 ‘é@/“%G B) 25 ‘é@/“%G  
C) 30 ‘é@/“%G D) 35 ‘é@/“%G

**Q.193.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r Trigonometry] Trigonometric Heights & Distances ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 1 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.194.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r ÖöFW n Math] Probability & Combinatorics Basics ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 2 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.195.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r ÖVç7W ation 3D] Circular Cylinder & Cone Surface Area ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 3 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.196.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r Time, Speed & Distance] Boat and Stream Relative Velocity ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 4 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.197.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r V G atic Equations] Algebraic Factorization & Polynomials ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 5 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.198.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r V G atic Equations] Algebraic Factorization & Polynomials ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 10 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.199.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r Trigonometry] Trigonometric Heights & Distances ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 11 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.200.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r ÖöFW n Math] Probability & Combinatorics Basics ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 12 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.201.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r ÖVç7W ation 3D] Circular Cylinder & Cone Surface Area ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 13 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.202.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r Time, Speed & Distance] Boat and Stream Relative Velocity ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 14 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.203.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r V G atic Equations] Algebraic Factorization & Polynomials ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 15 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.204.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r Trigonometry] Trigonometric Heights & Distances ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 16 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.205.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r ÖöFW n Math] Probability & Combinatorics Basics ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 17 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

**Q.206.** [“% ‘iM’ù>’iM’é •Y–“ù ”Ù7’és\$“â r ÖVç7W ation 3D] Circular Cylinder & Cone Surface Area ‘©0 •i““é0“ù\$ ‘©M“ 6”Ù( 18 ‘Y> ‘é>, ”Ù/“â 9”

- A) 24 B) 36  
C) 48 D) 60

- Q.207.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r Time, Speed & Distance] Boat and Stream Relative Velocity '©0 •i'“é0“ù\$ '©M“ 6"Û( 19 'Y> 'é>'  
A) 24 B) 36  
C) 48 D) 60
- Q.208.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r V G atic Equations] Algebraic Factorization & Polynomials '©0 •i'“é0“ù\$ '©M“ 6"Û( 20 'Y> 'é>'  
A) 24 B) 36  
C) 48 D) 60
- Q.209.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r Trigonometry] Trigonometric Heights & Distances '©0 •i'“é0“ù\$ '©M“ 6"Û( 21 'Y> 'é>', "Û/"â 9'  
A) 24 B) 36  
C) 48 D) 60
- Q.210.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r ÖöFW n Math] Probability & Combinatorics Basics '©0 •i'“é0“ù\$ '©M“ 6"Û( 22 'Y> 'é>', "Û/"â 9'  
A) 24 B) 36  
C) 48 D) 60
- Q.211.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r ÖVç7W ation 3D] Circular Cylinder & Cone Surface Area '©0 •i'“é0“ù\$ '©M“ 6"Û( 23 'Y> 'é>'  
A) 24 B) 36  
C) 48 D) 60
- Q.212.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r Time, Speed & Distance] Boat and Stream Relative Velocity '©0 •i'“é0“ù\$ '©M“ 6"Û( 24 'Y> 'é>'  
A) 24 B) 36  
C) 48 D) 60
- Q.213.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r V G atic Equations] Algebraic Factorization & Polynomials '©0 •i'“é0“ù\$ '©M“ 6"Û( 25 'Y> 'é>'  
A) 24 B) 36  
C) 48 D) 60
- Q.214.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r Trigonometry] Trigonometric Heights & Distances '©0 •i'“é0“ù\$ '©M“ 6"Û( 26 'Y> 'é>', "Û/"â 9'  
A) 24 B) 36  
C) 48 D) 60
- Q.215.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r ÖöFW n Math] Probability & Combinatorics Basics '©0 •i'“é0“ù\$ '©M“ 6"Û( 27 'Y> 'é>', "Û/"â 9'  
A) 24 B) 36  
C) 48 D) 60
- Q.216.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r ÖVç7W ation 3D] Circular Cylinder & Cone Surface Area '©0 •i'“é0“ù\$ '©M“ 6"Û( 28 'Y> 'é>'  
A) 24 B) 36  
C) 48 D) 60
- Q.217.** ["% 'iM'ù>'IM'É •Y-“ù "Û7'É\$“â r Time, Speed & Distance] Boat and Stream Relative Velocity '©0 •i'“é0“ù\$ '©M“ 6"Û( 29 'Y> 'é>'  
A) 24 B) 36  
C) 48 D) 60
- Q.218.** [•™\$“Û\$“ \*”Û0'iG“b \*” 2“ù8 'Y>•)8”Û "y,““ c Ã#CB \*” (“ M'©0” ”Û7“â \*”Û0“iM', EÒ 8•) ”Û/"é\$”Û.‘R ”Û7'É\$“â “ ,”Û/"é /'©M“ \$”  
A) “Y?’Y2”Û\* 1 “%09” 9”€ B) “Y?’Y2”Û\* 2 “Y?’y?’R 5 '©>' M'ù ”Û0'â '‰A“‰>“ \*”)0”Û#‘I “‰\$”Û/ “™H  
C) 'iK'‰K•“ “)\$ “™H• D) •y('éG•“ 8”r ”1 '‰09”
- Q.219.** [“‰ ‘, 2”1 “‰G“Y> •i/”1 •ù(' @•ò '9?’B \*”Û0“iM'‰\*iM“ \*”Û0“iM', Ò “)( (Calculus), “‰&“ù6 'É@'É '9?’B 'Y5“â “YM'ùB““ \*” ‘  
A) •Y5'Y2', .“é( = 0 B) “Y?’iM“)G“y#“é\$”Û.‘R 9”” Ö  
C) “‰.“é “)( “‰@'é> = pi / 2 D) 'é>'©>•) = " 2
- Q.220.** [“‰ ‘, 2”1 “‰G“Y> •i/”1 •ù(' @•ò '9?’B \*”Û0“iM'‰\*iM“ \*”Û0“iM', %Ò “)( (Calculus), “‰&“ù6 'É@'É '9?’B 'Y5“â “YM'ùB““ \*” ‘  
A) •Y5'Y2', .“é( = 0 B) “Y?’iM“)G“y#“é\$”Û.‘R 9”” Ö  
C) “‰.“é “)( “‰@'é> = pi / 2 D) 'é>'©>•) = " 2
- Q.221.** [“‰ ‘, 2”1 “‰G“Y> •i/”1 •ù(' @•ò '9?’B \*”Û0“iM'‰\*iM“ \*”Û0“iM', 5Ò “)( (Calculus), “‰&“ù6 'É@'É '9?’B 'Y5“â “YM'ùB““ \*” ‘  
A) •Y5'Y2', .“é( = 0 B) “Y?’iM“)G“y#“é\$”Û.‘R 9”” Ö  
C) “‰.“é “)( “‰@'é> = pi / 2 D) 'é>'©>•) = " 2
- Q.222.** [“‰ ‘, 2”1 “‰G“Y> •i/”1 •ù(' @•ò '9?’B \*”Û0“iM'‰\*iM“ \*”Û0“iM', EÒ “)( (Calculus), “‰&“ù6 'É@'É '9?’B 'Y5“â “YM'ùB““ \*” ‘  
A) •Y5'Y2', .“é( = 0 B) “Y?’iM“)G“y#“é\$”Û.‘R 9”” Ö  
C) “‰.“é “)( “‰@'é> = pi / 2 D) 'é>'©>•) = " 2

[illegible]

**Q.239.** [‰ ‘ , 2”1 ‹‰G“Y> •I/”1 •ù(‘ @•ò ’9?’B \*”Ù0“iM’‰\*”IM“ \*\*Ù0“iM’, # Ò ))( Calculus), “‰&“ù6 ’É@‘É ’9?’B ’Y5“â “YM”ùB“” \*  
A) •Y5‘Y2’, .‘é( = 0  
C) “‰.“é “)( “‰@‘é> = pi / 2  
B) “Y?“iM”)G“y#“és\$”Ù.‘R 9“” Ò  
D) ’é>’©>) = " 2

**Q.240.** [‰ ‘ , 2”1 ‹‰G“Y> •I/”1 •ù(‘ @•ò ’9?’B \*”Ù0“iM’‰\*”IM“ \*\*Ù0“iM’, #Ò Ò ))( Calculus), “‰&“ù6 ’É@‘É ’9?’B ’Y5“â “YM”ùB“” \*  
A) •Y5‘Y2’, .‘é( = 0  
C) “‰.“é “)( “‰@‘é> = pi / 2  
B) “Y?“iM”)G“y#“és\$”Ù.‘R 9“” Ò  
D) ’é>’©>) = " 2

**Q.241.** [‰ ‘ , 2”1 ‹‰G“Y> •I/”1 •ù(‘ @•ò ’9?’B \*”Ù0“iM’‰\*”IM“ \*\*Ù0“iM’, #5Ò “)( Calculus), “‰&“ù6 ’É@‘É ’9?’B ’Y5“â “YM”ùB“” \*  
A) •Y5‘Y2’, .‘é( = 0  
C) “‰.“é “)( “‰@‘é> = pi / 2  
B) “Y?“iM”)G“y#“és\$”Ù.‘R 9“” Ò  
D) ’é>’©>) = " 2

**Q.242.** [‰ ‘ , 2”1 ‹‰G“Y> •I/”1 •ù(‘ @•ò ’9?’B \*”Ù0“iM’‰\*”IM“ \*\*Ù0“iM’, #EÒ “)( Calculus), “‰&“ù6 ’É@‘É ’9?’B ’Y5“â “YM”ùB“” \*  
A) •Y5‘Y2’, .‘é( = 0  
C) “‰.“é “)( “‰@‘é> = pi / 2  
B) “Y?“iM”)G“y#“és\$”Ù.‘R 9“” Ò  
D) ’é>’©>) = " 2

**Q.243.** [‰ ‘ , 2”1 ‹‰G“Y> •I/”1 •ù(‘ @•ò ’9?’B \*”Ù0“iM’‰\*”IM“ \*\*Ù0“iM’, #UÒ “)( Calculus), “‰&“ù6 ’É@‘É ’9?’B ’Y5“â “YM”ùB“” \*  
A) •Y5‘Y2’, .‘é( = 0  
C) “‰.“é “)( “‰@‘é> = pi / 2  
B) “Y?“iM”)G“y#“és\$”Ù.‘R 9“” Ò  
D) ’é>’©>) = " 2